

Musterlösung Serie 12

RANK AND INVERTIBILITY

1. Rank. Suppose that V, W are finite-dimensional vector spaces over a field K . Let $T \in \text{Hom}(V, W)$. Prove that $\text{rank}(T) = 1$ if and only if there exists a basis $\mathcal{B}$ of V and a basis $\mathcal{C}$ of W such that with respect to these bases, all entries of $[T]_{\mathcal{C}}^{\mathcal{B}}$ equal 1.

Solution: We write m for the dimension of V and n for the dimension of W , and we denote by $\mathcal{B}_0$, respectively $\mathcal{C}_0$, an arbitrary basis of V , respectively W . Since T is a homomorphism of rank 1, the matrix $[T]_{\mathcal{C}_0}^{\mathcal{B}_0}$ is of rank 1. It is therefore equivalent to any matrix of rank 1 in $M_{n \times m}(K)$, namely there exist matrices $P \in \text{GL}_n(K)$ and $Q \in \text{GL}_m(K)$ such that $P[T]_{\mathcal{C}_0}^{\mathcal{B}_0}Q = A$, where $A \in M_{n \times m}(K)$ is the matrix whose entries are all equal to 1. Since $Q \in \text{GL}_m(K)$, it can be seen as the base change matrix from $\mathcal{B}$ to $\mathcal{B}_0$, where $\mathcal{B}$ is defined such that the diagram

$$\begin{array}{ccc} V & \xrightarrow{\text{Id}_V} & V \\ \downarrow \Phi_{\mathcal{B}} & & \downarrow \Phi_{\mathcal{B}_0} \\ K^m & \xrightarrow{P} & K^m \end{array}$$

commutes. Similarly, C defines a base-change matrix from $\mathcal{C}_0$ to some basis $\mathcal{C}$. Hence,

$$A = P[T]_{\mathcal{C}_0}^{\mathcal{B}_0}Q = [\text{Id}_W]_{\mathcal{C}}^{\mathcal{C}_0}[T]_{\mathcal{C}_0}^{\mathcal{B}_0}[\text{Id}_V]_{\mathcal{B}_0}^{\mathcal{B}} = [T]_{\mathcal{C}}^{\mathcal{B}}.$$

2. Invertible matrices. Suppose V is finite-dimensional and $S, T, U \in \text{Hom}(V, V)$ with $STU = \text{Id}_V$. Show that T is invertible and that $T^{-1} = US$.

Solution: Observe that $STU = \text{Id}_V$ implies that STU is an automorphism of V . Since $\{0\} = \text{Ker}(\text{Id}_V) = \text{Ker}(STU) \supseteq \text{Ker}(U)$, we must have that U is injective. Since V is finite-dimensional, this implies that U is bijective and hence invertible. So,

$$ST = U^{-1}.$$

Also since $V = \text{Im}(STU) \subseteq \text{Im}(S)$, we deduce that S is surjective. Since $S : V \rightarrow V$ and V is finite-dimensional, this implies that S is injective. Hence S is invertible and

$$T = S^{-1}U^{-1}.$$

This shows both that T is invertible and that

$$T^{-1} = US.$$

3. Invertible matrices. Let V be a finite-dimensional vector spaces over a field K . Suppose that $T \in \text{Hom}(V, V)$, and that $\mathcal{A} = \{u_1, \dots, u_n\}$ and $\mathcal{B} = \{v_1, \dots, v_n\}$ are bases of V . Prove that the following are equivalent:

- (a) T is invertible.
- (b) The columns of $[T]_{\mathcal{B}}^{\mathcal{A}}$ are linearly independent in K_{col}^n ;
- (c) The columns of $[T]_{\mathcal{B}}^{\mathcal{A}}$ span K_{col}^n ;
- (d) The rows of $[T]_{\mathcal{B}}^{\mathcal{A}}$ are linearly independent in K_{row}^n ;
- (e) The rows of $[T]_{\mathcal{B}}^{\mathcal{A}}$ span K_{row}^n .

Solution: See the proof of proposition 3.6.8 (p.128) in Menny Aka's lecture notes, available on the course website.

4. Morphisms of vector spaces. Let V be a finite-dimensional vector space. Prove or disprove:

- (a) Let $V' \subset V$ be a linear subspace. Every automorphism $f : V' \rightarrow V'$ can be extended to an automorphism $\bar{f} : V \rightarrow V$.
- (b) For every Endomorphism $f : V \rightarrow V$ its image $\text{Im}(f)$ is a linear complement $\text{Ker}(f)$ in V .
- (c) There does not exist any linear map $T : \mathbb{R}^5 \rightarrow \mathbb{R}^5$ such that

$$\text{rank}(T) = \dim \text{Ker}(T).$$

Lösung:

- (a) This assertion is true. Choose a complement of V' , i.e., a subvectorspace V'' of V with $V = V' + V''$ and $V' \cap V'' = \{0\}$. Then the map

$$V' \times V'' \rightarrow V, (v', v'') \mapsto v' + v''$$

is bijective. Define a map $\bar{f} : V \rightarrow V$ by $\bar{f}(v' + v'') := f(v') + v''$ for all $v' \in V'$ and $v'' \in V''$. As f is linear, a direct computation shows that $\bar{f}$ is linear as well. We claim that $\bar{f}$ is bijective. To see this, consider first $v' \in V'$ and $v'' \in V''$ with $f(v') + v'' = 0$. Then we have $f(v') = -v'' \in V' \cap V'' = \{0\}$ and so $f(v') = v'' = 0$. As f is injective, we also get $v' = 0$. Thus, we have $\text{Ker}(\bar{f}) = \{0\}$ and hence $\bar{f}$ is injective.

Now let $v \in V$ be an arbitrary element. Write $v = v' + v''$ with $v' \in V'$ and $v'' \in V''$. Then, we have $v' = f(f^{-1}(v'))$ and thus $v = f(f^{-1}(v')) + v'' = \bar{f}(f^{-1}(v') + v'')$. Therefore, the map $\bar{f}$ is surjective. Together we get that $\bar{f}$ is bijective and thus an isomorphism $V \rightarrow V$, and hence an Automorphism of V .

- (b) The assertion is false. Consider for example the endomorphism $L_A : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by left multiplication with

$$A := \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

Then, we have $\text{Ker}(L_A) = \langle (1, 0)^T \rangle = \text{Im}(L_A)$. The subspaces neither have intersection $\{0\}$, nor do they generate $\mathbb{R}^2$, and thus they cannot form $V = \text{Ker}(f) \oplus \text{Im}(f)$.

- (c) Consider an endomorphism $T : \mathbb{R}^5 \rightarrow \mathbb{R}^5$. By the rank theorem,

$$\text{rank}(T) + \dim(\text{Ker}(T)) = \dim(\mathbb{R}^5) = 5.$$

If we were to have $\text{rank}(T) = \dim(\text{Ker}(T))$, we would have

$$5 = 2\text{rank}(T) \in 2\mathbb{Z},$$

which cannot hold since 5 is not even.

5. Powers of matrices. Consider the space $M_{2 \times 2}(K)$ of 2-by-2 matrices over a field K .

- (a) Show that if $A \in M_{2 \times 2}(K)$ satisfies $A^2 \neq 0$, then $A^k \neq 0$ for all $k \geq 3$.
 (b) Find a field K and a matrix $A \in M_{2 \times 2}(K) \setminus \{0\}$ such that $\exists n \in \mathbb{N} : A^n = 0$.

Solution:

- (a) Let $A \in M_{2 \times 2}(K)$ and denote $T = T_A$ be the linear map associated to A . If $T(K^2) = K^2$, then T is an isomorphism and $T^k \neq 0$ for all $k \geq 1$. Assume now that $T(K^2) = L$ is a one-dimensional subspace of K^2 . Then $T^2(K^2) = T(L) \subseteq L$. However, since $T^2 \neq 0$, $T^2(K^2) = T(L) = L$.

Let $n \geq 2$. We now assume that $T^m(K^2) = L$ for $2 \leq m \leq n$ and we show that $T^{n+1}(K^2) = L$. Indeed,

$$T^{n+1}(K^2) = T(T^n(K^2)) = T(L) = L.$$

This shows that $T^k \neq 0$ for all $k \geq 1$.

- (b) Take for example

$$\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \in M_{2 \times 2}(\mathbb{C}).$$

6. Rank of matrices. Determine the ranks of the following rational $n \times n$ -matrixes, those being elements of $M_{n \times n}(\mathbb{Q})$, depending on the positive integer n .

- (a) $(kl)_{k,l=1,\dots,n}$;
- (b) $((-1)^{k+l}(k+l-1))_{k,l=1,\dots,n}$;
- (c) $\left(\frac{(k+l)!}{k!l!}\right)_{k,l=0,\dots,n-1}$.

Hint: Note that the last matrix is indexed from 0 to $n-1$. You may use induction to find the desired formula.

Solution:

- (a) Define $B := (b_{kl})_{k,l=1,\dots,n} := (kl)_{k,l=1,\dots,n}$. Let $B' = (b'_{kl})_{k,l=1,\dots,n}$ be the matrix arising from B by subtracting for every $k = 2, \dots, n$ the k -multiple of the first row from the k -st row. Then, we have

$$b'_{kl} = \begin{cases} b_{kl} = l & \text{falls } k = 1 \\ b_{kl} - kb_{1l} = kl - kl = 0 & \text{falls } k > 1. \end{cases}$$

Thus, the matrix B' has exactly one non-vanishing row and thus has rank 1. As B' arose through elementary row operations from B , and hence by left multiplication with an invertible matrix, the rank of B' is the same as the rank of B . Therefore, we get $\text{Rang}(B) = 1$.

Aliter: Let $u := (1, \dots, n)$ the $1 \times n$ matrix with entry i on the position $(1, i)$. Then, we have $B = u^T \cdot u$. As the rank of (u) is ≤ 1 , exercise 3 (c) yields $\text{Rang}(B) \leq 1$. From $B \neq 0$, we also get $\text{Rang}(B) \geq 1$ and thus $\text{Rang}(B) = 1$.

- (b) Let $B := (b_{kl})_{k,l=1,\dots,n}$ with $b_{kl} := (-1)^{k+l}(k+l-1)$. For $n = 1$, the matrix $B = (1) \neq 0$ has rank 1. It remains to treat the case $n \geq 2$. For all $k = 1, \dots, n-2$ and $l = 1, \dots, n$, we get

$$b_{kl} + 2b_{k+1,l} + b_{k+2,l} = 0,$$

and thus the k -th row of B is a linear combination of the $(k+1)$ -th and $(k+2)$ -th row. Therefore, the matrix B can by elementary row operations be transformed into a matrix, in which only the last two rows are non-vanishing and these are identical to the last two rows of B . These two are linearly independent, which we get from a direct computation. Together, we have

$$\text{rank}(B) = \begin{cases} 1, & \text{falls } n = 1 \\ 2, & \text{falls } n \geq 2 \end{cases}$$

(c) Let $C_n := (c_{kl})_{k,l=0,\dots,n-1}$ be the matrix with $c_{kl} := \frac{(k+l)!}{k!\ell!} = \binom{k+l}{l}$.

Claim: $\text{rank}(C_n) = n$.

Proof: We induct over n . For $n = 1$, the assertion is true, as $C_1 = (1) \neq 0$. Assume, that we know the claim for $n \geq 1$. Let $C' = (c'_{kl})_{k,l=0,\dots,n}$ be the matrix, which arises from C_{n+1} , when beginning with the last row, from each row the previous row is subtracted. In other words:

$$c'_{kl} := \begin{cases} c_{kl} - c_{k-1,l} & \text{if } k = 1, \dots, n \\ c_{0l} & \text{if } k = 0. \end{cases}$$

Let moreover $C'' = (c''_{kl})_{k,l=0,\dots,n}$ be the matrix, which arises from C' , when beginning with the last column, from each column the previous column is subtracted. In other words:

$$c''_{kl} := \begin{cases} c'_{kl} - c'_{k,l-1} & l = 1, \dots, n \\ c'_{k0} & l = 0. \end{cases}$$

For every $1 \leq k, l \leq n$, we have

$$\begin{aligned} c''_{kl} &= c'_{kl} - c'_{k,l-1} \\ &= (c_{kl} - c_{k-1,l}) - (c_{k,l-1} - c_{k-1,l-1}) \\ &= \frac{(k+l)!}{k!\ell!} - \frac{(k+l-1)!}{(k-1)!\ell!} - \frac{(k+l-1)!}{k!(\ell-1)!} + \frac{(k+l-2)!}{(k-1)!(\ell-1)!} \\ &= \frac{(k+l)!}{k!\ell!} \left(1 - \frac{k}{k+1} - \frac{l}{k+l} \right) + \frac{(k+l-2)!}{(k-1)!(\ell-1)!} \\ &= \frac{(k+l-2)!}{(k-1)!(\ell-1)!}. \end{aligned}$$

Therefore, we get

$$C'' = \begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 & & & \\ \vdots & & C_n & \\ 0 & & & \end{pmatrix}$$

and finally

$$\text{Rang } C_{n+1} = \text{Rang}(C'') = 1 + \text{Rang } C_n = n + 1.$$